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Routeing Road Sweepers in a Rural Area

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A computer program has been developed to assist the routeing of road-sweeping vehicles in a part of the north-west of England. The aim of the study was to develop a method which would specify the route to be followed by each sweeper which minimized the dead distance travelled, subject to constraints on the capacity of the sweeper and the time available in any day. The method had to be flexible so that the effects of changes, such as new vehicles, alterations to the location of tips for detritus, or different frequencies of sweeping could be examined quickly. The paper describes how the design of the heuristic algorithm was influenced by the geography of the area, the computing facilities and the time available to complete the study.

Key words: Chinese postman problem, heuristics, local government, microcomputers, routeing

BACKGROUND

Highway authorities have a duty to sweep roads for public safety and for maintenance. This must be done at a requisite minimum frequency. In addition, the local authority has discretion to sweep roads more frequently for amenity and public-health reasons. The local authority may carry out the sweeping using their own men and vehicles, but may agree to have certain areas swept by other local authorities. Since the introduction of the Local Government Planning and Land Act 1980, the direct-labour organizations employed by the local authorities must demonstrate good value for money and must compete with outside contractors, by way of tender, for a proportion of their work loads, which could include road sweeping.

Lancashire County Council found that their manual method of routeing road-sweeping vehicles was too slow and inflexible to take account of changes which may take place in the road-sweeping requirements. They therefore asked a team from the Department of Operational Research and Operations Management at Lancaster University to develop a method for routeing their road-sweeping vehicles. The new method had to be efficient in that it aimed to minimize the dead distance travelled (i.e. the roads which had to be traversed, without being swept). It was also necessary for the method to be sufficiently precise so that the work done could be measured. This would be needed when bidding for the sweeping of a new area or assessing bids from other contractors to ensure good value for money. The measurement of the work was also important as a component of the drivers' bonus payments.

The method used was to be sufficiently flexible to be able to incorporate new roads, new vehicles with different speeds or capacities and new policies on the frequency of road sweeping, and to have the ability to incorporate local knowledge. A secondary objective was that management preferred a routeing system which ensured that both sides of a road were planned to be swept in the same day. This would make checking and supervision of the work easier.

These requirements pointed to designing a computer-based tool. In order for it to be used by area level management, it needed to be capable of running on a PC-compatible machine, several of which were locally available.

DESCRIPTION OF THE PROBLEM

The county is divided into several districts. Each district is swept by a sweeper based at a local depot. The sweeper starts from the depot at the start of the day and must return there empty at the end of the day. The limited capacity of the road-sweeping vehicles means that a sweeper must be emptied perhaps several times during the day as well as at the end of the day's sweeping. One feature of the rural situation compared with many similar problems in urban areas is that there are several tipping sites from which to choose to empty the sweeper, which are distributed round the district. The depot is not generally a tipping site.

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The roads in a district are classified according to the frequency with which they are to be swept. For example, 'A' roads might be swept at least 12 times a year while some rural roads might be swept only once.

There are two further features of the type of situation examined here which would not necessarily be found in a more urban environment. The first is that there are very few one-way streets in the area. In the few cases where one-way streets exist, they are normally swept in one direction by a sweeper going against the traffic taking suitable precautions, so they could be regarded as two-way streets from the point of view of forming routes. Dual carriageways, where both the nearside and offside kerbs are to be swept, can be dealt with by simply representing the dual carriageway as two roads between the same pair of junctions. The second feature is that there are no additional constraints which imply that certain roads can only be swept at certain times. This is sometimes the case in urban environments where parking regulations may have a significant effect on the ability of sweepers to sweep the kerbs and thus require that the roadsides are further classified according to their availability for sweeping in different time windows; see e.g. Bodin and Kursh.[1,2]

SOLUTION APPROACH

The problem of finding a route of minimum distance which covers all arcs in a network is known as the Chinese postman problem. The routeing problem described in the previous section is a directed Chinese postman problem with constraints on the capacity of the vehicles and the time available to them in a day. Although there are efficient algorithms available for solving the directed and undirected Chinese postman problems optimally, the addition of capacity constraints makes the problems much harder. Golden and Wong[3] have shown that the capacitated Chinese postman problem (CCPP) is NP-hard, which means that it is one of a set of many combinatorial optimization problems for which no algorithm has yet been devised which will guarantee an optimal solution in a number of steps which is a polynomial function of the size of the problem. Other practical routeing problems, such as refuse collection and winter gritting operations, can also be formulated as a CCPP, but in these examples the vehicles only have to pass down each road in the network once. These are therefore examples of undirected CCPPs and methods suggested for solving problems of this type (e.g. methods described in Golden *et al.*[4]) are not appropriate to the road sweeping problem. A discussion of approaches to various arc-routeing problems is contained in Bodin *et al.*[5]

Therefore it was decided to design a heuristic to produce feasible routes, with as little dead distance as possible, though without guaranteeing an optimal solution. This is the approach adopted by Bodin and Kursh.[1,2] However, Bodin and Kursh's method is designed to cope with an area containing many one-way streets, as are found in New York. Therefore it was inappropriate to make use of their method for an area where all roads could be treated as two-way streets. Treating all roads as pairs of one-way streets would imply extra data storage requirements and use of Bodin and Kursh's method would not generate routes which implied that both sides of any road would be swept on the same day. Bodin and Kursh's method also deals with time window constraints, an unnecessary complication in our application. Indeed, taking advantage of the fact that all roads could be treated as two-way streets meant that it was possible to devise a simple algorithm which could easily incorporate user interaction if necessary and would generate routes which implied that both sides of any road would be swept on the same day.

It was decided to design the heuristic within a visual interactive model so that users could test the consequences of alternative routeing decisions as well as obtaining results for different scenarios.

Two research assistants were employed for a year to work on this project plus two other projects requiring a similar amount of work for the same client. They did not have the benefit of being able to make use of a digitized road network database, as the County was still in the process of deciding which geographical information system to use and did not have such a database available. Therefore it was decided that one district would be selected on which the routeing program would operate and that part of the project would have to include digitizing the road network in this district. This was done by creating text files of the co-ordinates of road junctions and bends

using a digitizer in the Department of Geography at Lancaster University. The digitizer is a machine connected to a computer which allows a mouse to be moved across a map table. Whenever one of the mouse buttons is clicked, the digitizer records the map co-ordinates of the position of the mouse. This enables files to be constructed giving the co-ordinates of road junctions and road bends. A database had then to be constructed which showed which roads met at which junctions, and which recorded road lengths and the positions of the depot and tips. The district used contained 624 nodes representing junctions or ends of culs-de-sac (i.e. dead-ends) and 807 roads with 4 tipping sites (see Figure 1). The preparation and checking of the database took at least two person-months of work.

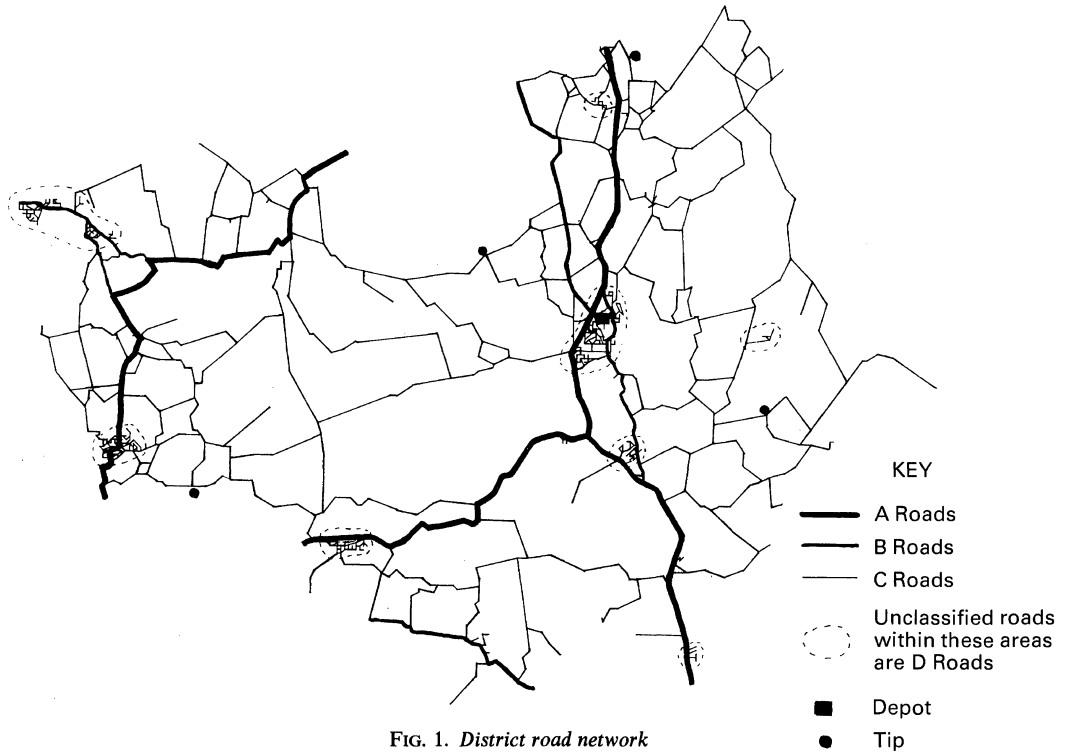


FIG. 1. *District road network*

The approach to this problem was partly influenced by the limited time available. The particular heuristic approach adopted was one where we were sure the basic algorithm could be implemented in the time available, but, if time allowed, the heuristic could be refined to reduce the occasions where the heuristic performed badly. As will be described in the next section, the team were able to develop the heuristic to an extent where it produced acceptable routes in most situations. By designing the heuristic within a visual interactive modelling environment, the system makes it easy for a user to modify the routes produced if he or she wishes to test alternatives. As a strong visual representation of the problem is available through the map, human abilities can be used to detect areas of possible improvement which could be difficult to program automatically. The interactive heuristic approach is discussed more generally and with reference to other examples in Eglese.[6]

SOLUTION METHOD

The problem is made more tractable by decomposing it in two ways: first on the basis of sweeping frequency and secondly according to the tip which will be used to empty the sweeper.

The roads in a district are classified according to the frequency with which they are to be swept. The classifications used in the study were:

A—'A' roads,
B—'B' roads,
C—unclassified rural roads,
D—unclassified estate or village roads.

The higher the class of road, the more often it was required to be swept, so the computer program allows four possible subsets of roads to be selected for a sweeping schedule:

1—A roads only,
2—A and B roads only,
3—A, B and C roads only,
4—A, B, C and D roads.

Then if, for example, A roads are to be swept 12 times in a year and B roads 8 times a year then the A road schedule (option 1) can be used 4 times and the A and B road schedule (option 2) can be used 8 times in a year. This decomposition allows changes to the frequency of sweeping different classes of roads to be made easily. In the example given, the proposed method implies that B roads will be swept irregularly, sometimes every month and sometimes every other month. If this had been unacceptable, it would have been necessary to prepare routes which swept B roads only.

Each road in the district is also classified according to the nearest tip. The program develops routes for each tip area separately, and so ensures that the nearest tip is always used when the sweeper must be emptied during the course of a day. Tests were carried out to see whether any improvement could be gained by splitting up the district, so that each tip area consisted of the roads for which the distance to the tip plus the distance to the depot was minimized, but this strategy increased the dead distance travelled.

The routeing algorithm is based on the observation that given a set of two-way roads which must be swept each side and no capacity constraints, a route may be constructed to sweep both sides of each road without incurring any dead distance by simply taking the next unswept road at each junction and returning down the other side of the present road if all other roads from the junction have been swept on one or both sides (see Figure 2). In urban areas this method of routeing may require modification if there are problems in carrying out the U-turns implied. However, in the predominantly rural setting of this study no modifications were needed to reduce the number of U-turns.

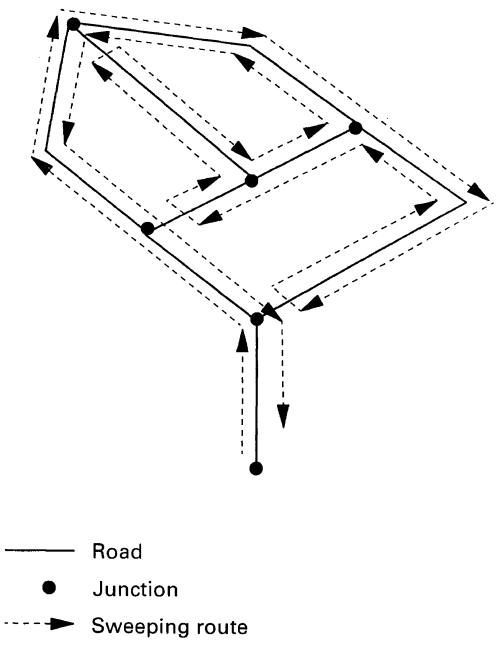


FIG. 2. *Illustration of the basic routeing method.*

The algorithm proceeds from one junction to a neighbouring junction at each iteration. As each junction is reached, a record is kept of the distance swept since the sweeper was last emptied and the time elapsed since the start of the day. A test is applied to see whether the distance swept since the sweeper was last emptied plus the distance to be swept down roads which have had only one side swept is greater than the distance which the sweeper can sweep before emptying. If this is so, the time available for sweeping is reduced by an estimate of the time required to go to the nearest

tip, empty the vehicle and return to continue sweeping. A second test calculates whether the time remaining in the day is enough to enable the sweeper to sweep the road being considered, finish sweeping the other sides of roads swept once, and finally empty and return to the depot. When the sweeper finally empties at the end of the day, it is routed to go to the tip for which the distance to the tip from the end of sweeping plus the distance from the tip to the depot is minimized. If neither test is positive, the algorithm then normally proceeds by selecting the next unswept road on the left as the road to be next swept, returning down the other side of the last road if no unswept road is available.

This method of producing routes has the advantage from a supervisory point of view that when a road is included in the route for the day, both sides of the road are due to be swept in the same day.

The first route constructed in a tip area is started at the junction nearest to the depot, though this may be overridden to any other junction if required. Each new day's route starts as close as possible to where the last day's route finished. This means that if a route is not completed one day it could be completed on the next day and the next day's route started without a long dead distance in between.

If the route returns to the starting junction and there are no more unswept roads off it, but there is still time remaining in the day, the algorithm will search for the nearest junction from which to continue sweeping. This is required, for example, when the roads to be swept form disconnected sets. Dijkstra's algorithm is used whenever shortest distances and routes are required. Smith[7] provides a suitable Pascal code.

After testing a version of the program as described above, it was clear that some of the routes produced were not particularly efficient in terms of the amount of dead distance travelled. In particular, the following situations were identified and are illustrated in Figure 3:

(a) If a road which was a dead-end happened to be on the right at a junction, it might get omitted from the current day's routes owing to the capacity and time constraints. The dead-end then became an isolated stretch of road which could be swept only after incurring extra dead distance.

(b) This situation involves estates or villages adjacent to a major road. Using the original method, the rule could leave only part of the estate or village swept before the sweeper is routed away down the major road.

(c) Some roads could be left isolated and unswept after the sweeping route had passed through the junctions at each end of the road.

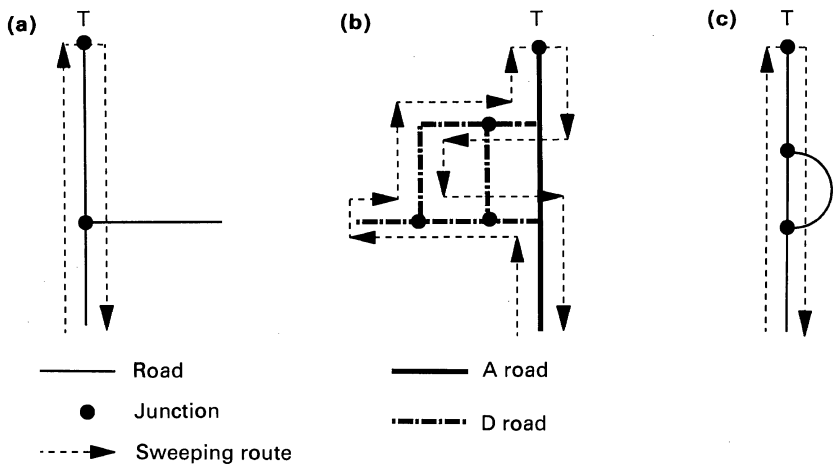


FIG 3. *In each diagram, junction T is the junction where the time constraint means that the sweeper must return to sweep any roads only swept on one side so far, before emptying and returning to the depot.*

These problems were overcome by modifying the original rule of sweeping the next junction on the left by the following set of rules, which give a priority order for selecting the next road to be swept:

1. Any road which is a dead-end.
2. Any category D roads to be swept.
3. Any road which has had the sweeping route pass through the junction at its other end. (This means it has effectively become like a dead-end.)
4. Otherwise, the next road on the left.

As before, if all roads are swept on at least one side, the sweeper returns down the other side of the road just swept.

This produced much more acceptable sweeping routes automatically. However, it was still clear that there were cases where better routes could be constructed, but the complexity of the selection procedure to find these improvements automatically was becoming such that the time required to develop the rules and run the program were excessive. The program therefore allows the user to override the above selection of the next road to be swept and to specify an unswept road of his or her choosing at any junction in the route. Figure 4 shows an example of a road network configuration where the automatic routeing may produce an unacceptable set of routes and where manual intervention could be used to overcome the problem.

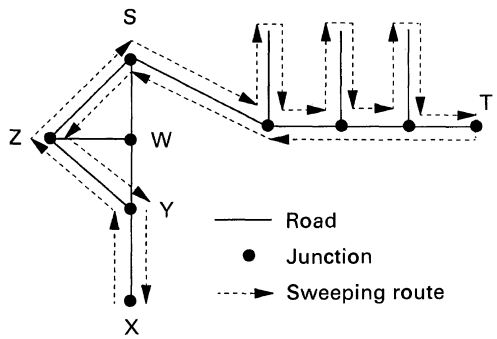


FIG. 4. *The sweeping route represents a day's sweeping, starting and finishing at X. The rules used leave SW, ZW and YW unswept and isolated from the rest of the network if this can only be reached via X or T. A better set of routes may result from directing the sweeper to go to W instead of Z from Y. Roads SW, ZW and YW will then be swept before leaving roads near junction T for the next day's route.*

In the proposed method for selecting the next road at a junction, no account is taken of how far the tip will be from the sweeper when the sweeper next needs emptying. It was decided not to attempt to take this into account because there could be variability in the distance swept before the sweeper was full, depending on the weather conditions when sweeping and the state of the roads. A routeing method which relied on the stability of the capacity constraint for its effectiveness would often be found to be impractical. If the distance swept before the sweeper was full were more consistent, it would have been possible to modify the proposed algorithm so that the capacity constraint was treated in a similar way to the time constraint. The sweeper would then start and therefore finish each section before emptying as close as possible to each tip.

Of course, the variability in the distance swept before the sweeper is full is a problem for any method used to plan the work. Some of the variability can be anticipated on a particular day; for example when wet dead leaves must be swept up the sweeper will become full relatively quickly. However, there is still much variability which cannot be anticipated in advance. Nevertheless, the computer system enables the sensitivity of the workload to this variability to be explored. The sweeping speed and speed covering dead distance are both relatively stable parameters, but the consequences of variations can be examined using the computer program.

USING THE COMPUTER PROGRAM

The program was written in Turbo Pascal and designed to run on a PC-compatible with a colour monitor and graphics capability.

The database is first read in and the user specifies which set of roads is required to be swept, which tip area is to be used, and from which junction the first route is to start. Before routeing

commences, the user has the opportunity of altering the following parameters from their initial default values:

the sweeping speed;
the dead travel speed;
the length of day;
the capacity of the sweeper (in terms of maximum distance before emptying);
the tip area to which a road is assigned;
the class of a road;
the priority of selecting a particular road for sweeping when constructing the routes.

The visual display shows a map of the tip area and the roads to be swept. As a road is swept on one side its colour is changed and when it has been swept on the other side the colour is changed again. The time spent sweeping, travelling over dead distance and in total are shown on the screen, as are the corresponding distances. A message indicates when and where the sweeper is full and must go to the tip to empty. There is also a facility to 'zoom' so that part of the area can be seen in greater detail. The user may also opt to use the manual override while the route is being constructed. In this mode, the road which the program would select as being the next to be swept is flashed on the screen and the user may accept or reject it. If it is rejected and a different road could be selected then the alternative roads are successively flashed until one is accepted.

A text file is created by the program which contains the details of the routes constructed and gives the total distances and times required. This file provides the basic information required to measure the work involved in sweeping an area and it may also be used to produce maps showing the different routes. The file can also be used to produce a written explanation of the routes for the driver of the sweeping vehicle. An example of this form of output is shown in Figure 5.

RESULTS AND CONCLUSIONS

When the computer program had been applied to the district under study which was swept by one sweeper, the routes formed implied that the sweeper could complete the sweeping required with the specified frequencies over a year in a total of 62 working days. On average the sweeper had to be emptied twice during each day's route plus once at the end of the day. The solution required some roads to be reassigned to different tip areas and in a few cases the automatic route selection was overridden. It was difficult to make a precise comparison between the solution generated using the computer program and current practice. This was because road sweeping had been managed in the past on a broad basis without the need for measurement. The sweeper was also used for additional work such as sweeping accident debris or attendance on surface dressing operations. However, the impression was that about half of the sweeper's working time was spent on routine sweeping. The result then clearly represented a reduction in the number of days that were estimated to be used by the sweeper at present on routine sweeping.

In practice some of the work additional to routine sweeping may take part of a day, leaving the rest for routine sweeping. The computer program provides a flexible tool which enables routes to be set up taking a shorter time than a full day if that is what is required.

Local management have used the program to produce routes for the district, sometimes adjusting the parameters used and have found the program easy to use. The program can be run without any user intervention to give routes for a set of roads in a district within a few minutes. The driver's instructions for some of the routes produced were taken and tested out in practice. The instructions were found to be clear to follow.

The County has now obtained a geographic information system and is well on the way to completing the digitization of the complete road network. Since the project was completed the County's programmers have written an interface which allows the roads in any area of the County to be selected and the information passed to a PC in a form which will input to the sweeping programme.

The software provided has been accepted as a useful tool for the management of the road-sweeping operation. The program ensures that feasible and efficient routes are produced, with the computer taking care of all the calculations of distances covered and time spent. The amount of manual intervention required to produce good routes is minimal. The program is flexible and easy

To get from the depot to the start of this route, the shortest way is as follows:

From the depot go down Green Lane (towards the junction with Croston Road, Preston Lancaster New Road (A6), Park Lane (B5377)).

Turn right into Park Lane (B5377) (second junction—320 yards).

Turn left into Broad Lane (second junction—0.6 miles).

Drive along Broad Lane for a further (630 yards) at which point you should have reached the start of the route.

Start sweeping New House Lane at the junction with Broad Lane, Nateby Bungalows Lane, Whitters Lane.

Turn left into Church Lane/New House Lane (0.8 miles) and continue sweeping.

Turn left into Church Lane (240 yards) and continue sweeping.

Carry on until tip on Church Lane (0.7 miles), do a U-turn, and continue sweeping back down the other side of Church Lane.

Turn left into School Lane (0.7 miles) and continue sweeping.

Turn left into Park Lane (B5377) (0.6 miles) and continue sweeping.

Carry on until the junction with Ratcliffe Lane (1.0 miles), do a U-turn, and continue sweeping back down the other side of Park Lane (B4377).

Turn left into School Lane (second junction—1.4 miles) and continue sweeping.

Turn left into Church Lane/New House Lane (0.6 miles) and continue sweeping.

Turn right into New House Lane (240 yards) and continue sweeping.

Continue sweeping for a further (0.8 miles) until the junction of New House Lane and Park Lane (B5377) is reached.

The shortest way to the tip is as follows:

From the present junction go down Clevely Bank Lane (towards the junction with Hollins Lane).

Turn left into Hollins Lane (380 yards).

Turn left into Preston Lancaster Road (A6) (620 yards).

Turn right into Cabus Nook Lane (second junction—0.5 miles).

Turn right into Broad Lane (0.9 miles).

Turn right into Church Lane/New House Lane (second junction—0.6 miles).

Turn left into Church Lane (240 yards).

Drive along Church Lane for a further (0.7 miles) at which point you should have reached the tip.

FIG. 5. *An example of the directions given to the sweeper driver. This example of part of a day's work gives directions from the depot to the start of sweeping in New House Lane, the route to follow while sweeping until the sweeper is first full and directions to the nearest tip in Church Lane, where the sweeper is emptied.*

to use, so alternative routes can be evaluated quickly. The program also enables investigations into the consequences of changes such as characteristics of new sweeping vehicles or changes in depot or tipping sites to be carried out with ease.

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